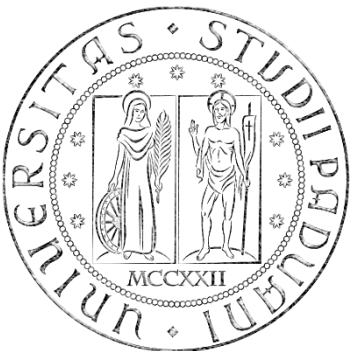


Biomedical Wearable Technologies
for Healthcare and Wellbeing

UX and UI in Flutter

A.Y. 2025-2026
Giacomo Cappon



User eXperience (UX)

- **UX** (User eXperience) is the **total experience** a person has while interacting with a product.
- It includes perception, emotional response, efficiency, clarity, friction, trust.
- UX is not just screens. UX is the entire journey.
- When UX is ok? You should ask yourself:
 - Does the product solve a real problem?
 - Is the user goal clear?
 - Is the path to that goal obvious?
 - Does the system behave predictably?
 - Is error handling supportive?



User Interface (UI)

- **UI** (User Interface) is the **visual and interactive layer**.
- It includes typography, color, layout, spacing, components, animations,...
- What is the difference between UX and UI?

UI communicates structure.
UX defines structure.



Fundamental UX Principles

- UX grounds on **five** principles:
 - Visual Hierarchy
 - Consistency
 - Feedback
 - Cognitive Load
 - Accessibility
- These apply whether you build a website, mobile app, medical device, or car dashboard.



Visual Hierarchy

➤ Users scan. They do not read linearly.

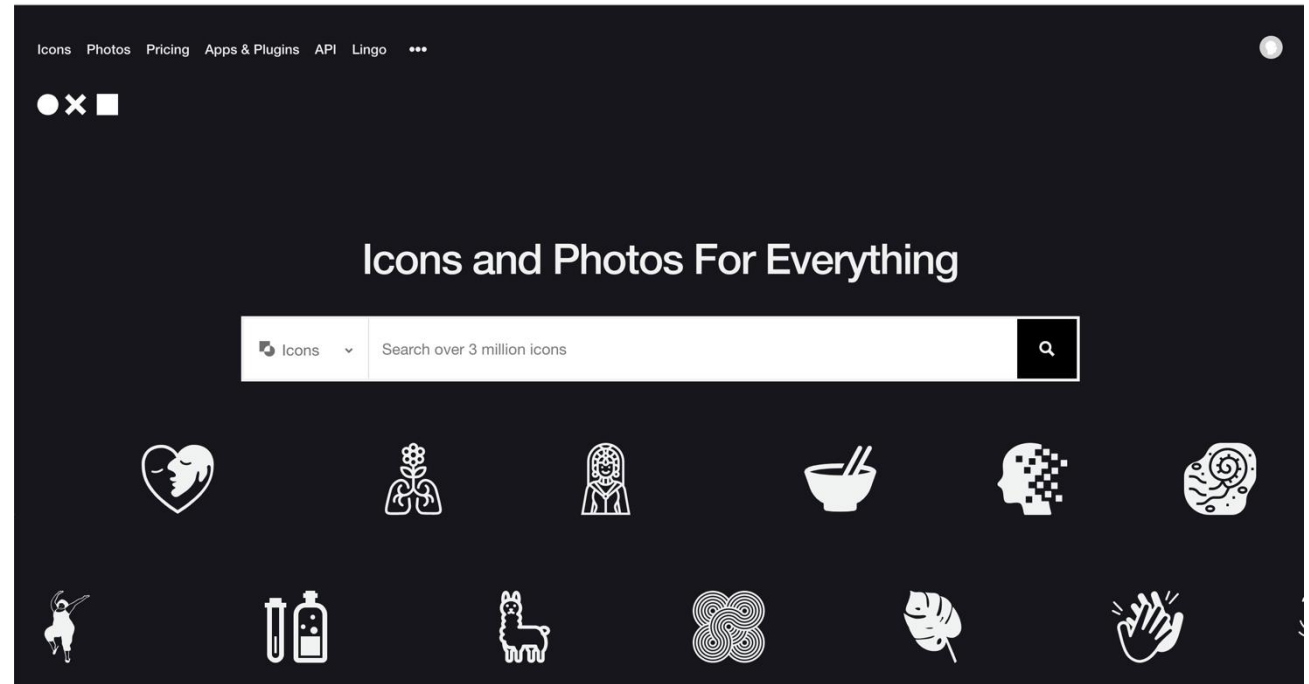
➤ Hierarchy answers:

- What matters most?
- What is actionable?
- What is secondary?

➤ Tools in our hands:

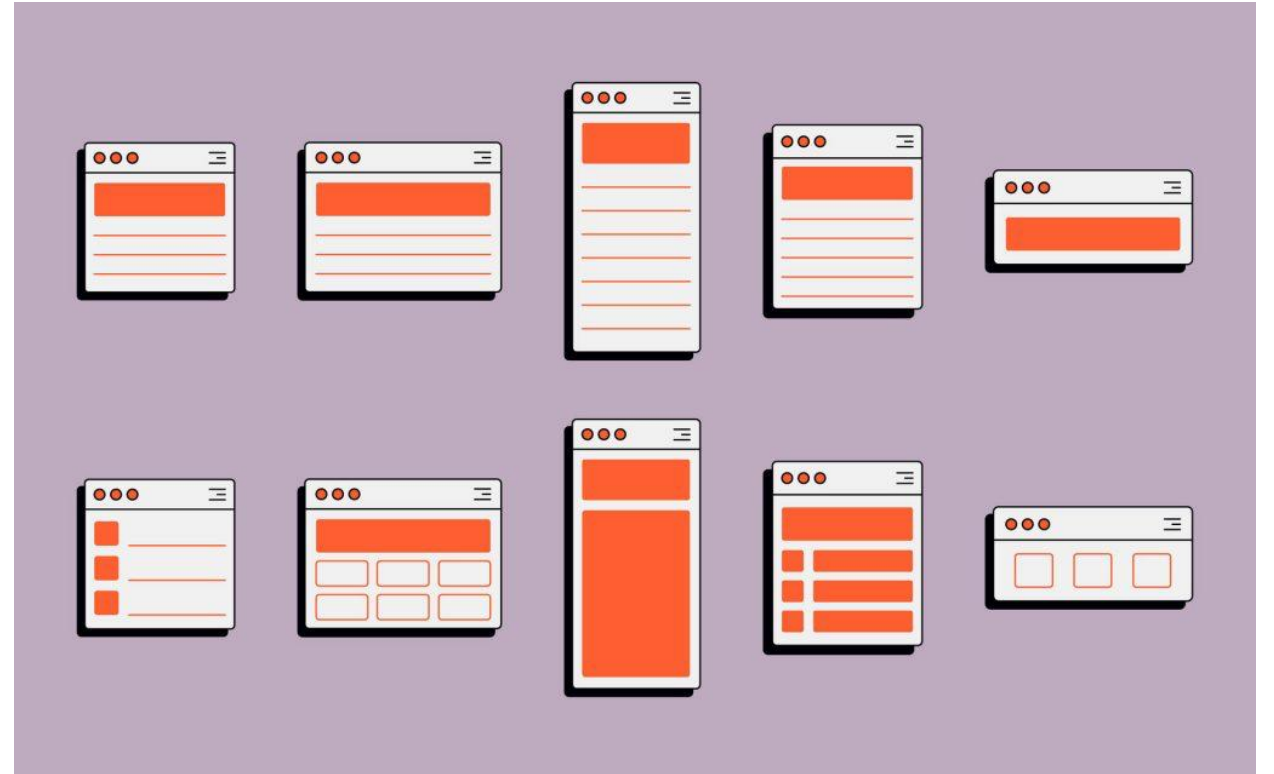
- Setting the size of each component
- Setting the contrast
- Defining spacing
- Setup and alignment policy
- Grouping stuff together

➤ If everything is bold, nothing is bold.



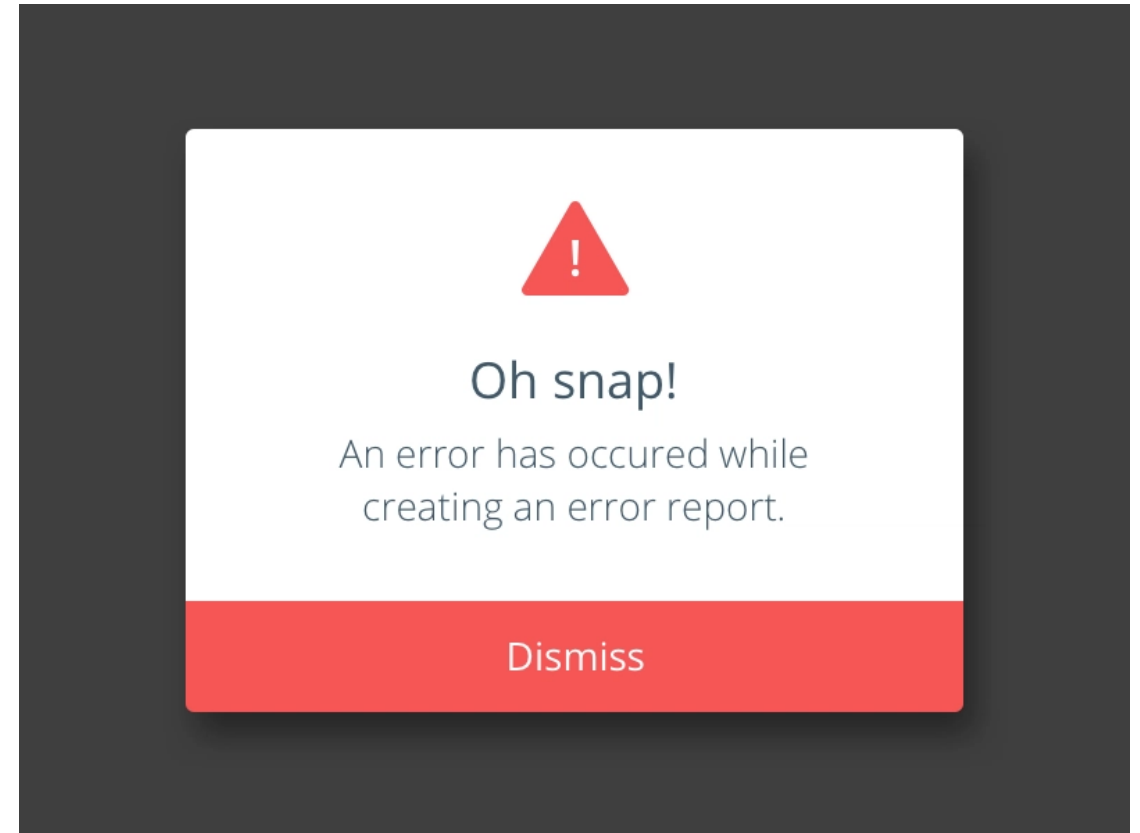
Consistency

- Humans recognize patterns.
- If
 - Buttons look different each time
 - Navigation changes unpredictably
 - Terminology variesthen, the cognitive load increases.
- Consistency builds confidence.



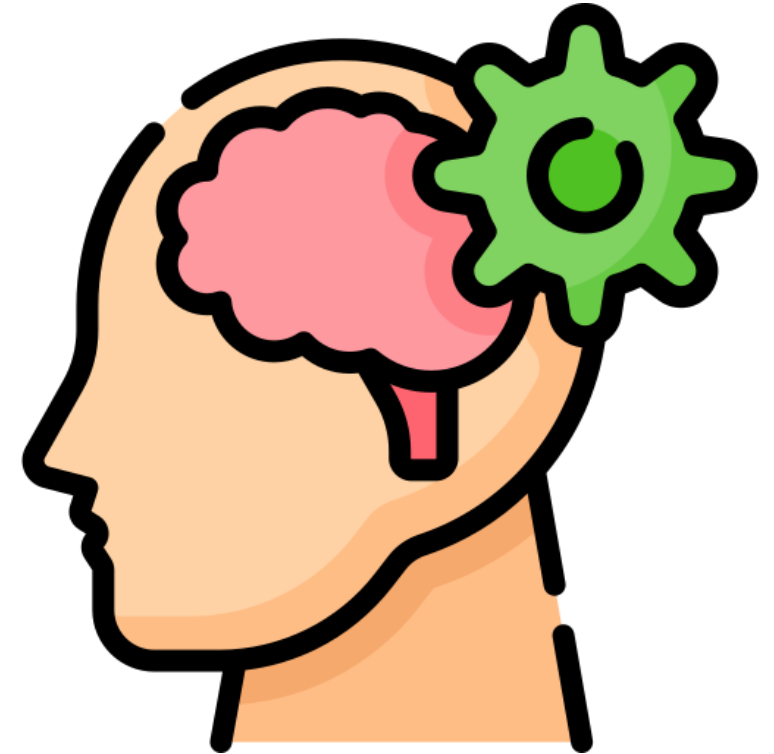
Feedback

- Every action must produce feedback.
- Examples:
 - Button press animation
 - Loading indicator
 - Error message
 - Success confirmation
- No feedback = anxiety. Did it worked?



Cognitive Load

- Cognitive load = Mental effort and stress
- Cognitive load increases when:
 - Too many options exist
 - Too much information is visible
 - Terminology is complex
 - Interaction patterns vary
- Goal: reduce thinking.



Accessibility

➤ Accessibility means:

- Readable contrast
- Scalable text
- Clear semantics
- Usable with assistive technologies

➤ Accessibility is not a feature. It is a quality standard.



Design Systems

- A **design system** is a comprehensive, centralized repository of reusable components, standards, documentation, and design principles used to create consistent digital products at scale.
- It prevents:
 - Random design decisions
 - Visual inconsistency
 - Scaling problems
- Examples of design systems:
 - Apple Human Interface Guidelines
 - Microsoft Fluent
 - Material Design

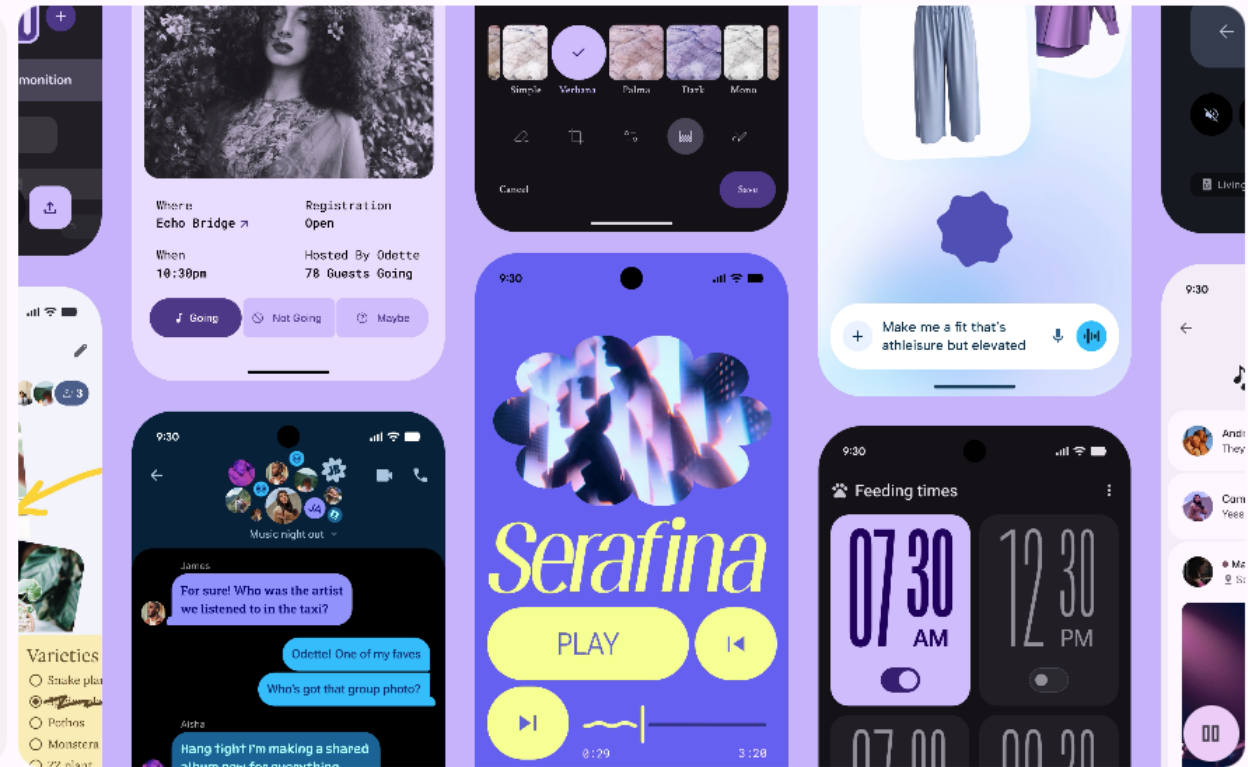


Material Design

Material Design

Material Design 3 is Google's open-source design system for building beautiful, usable products.

Get started



Material Design

- Material Design is not “Google style.”
- It is a **UX framework encoded into UI rules.**
- **Core ideas:**
 - *Structure through surfaces*: organizing UI into background, surfaces, and elevated components to create visual grouping.
 - *Meaningful motions*: explaining what changed, where content came from, and what happened.
 - *Clear hierarchy*: defining scales, button hierarchy, component roles.
 - *Color system*: using a primary, secondary, surface, error, contrasts
- This ensures accessible combinations.



Material Components as UX Decision

- Material components are structured by importance.
- **Let's make some examples**
- All examples and best practices can be found at:
<https://m3.material.io/>



Material Components as UX Decision

➤ Buttons:

- Primary action → Elevated, Filled, and Tonal button
- Secondary → Outlined button
- Tertiary → Text button

This encodes UX hierarchy visually.

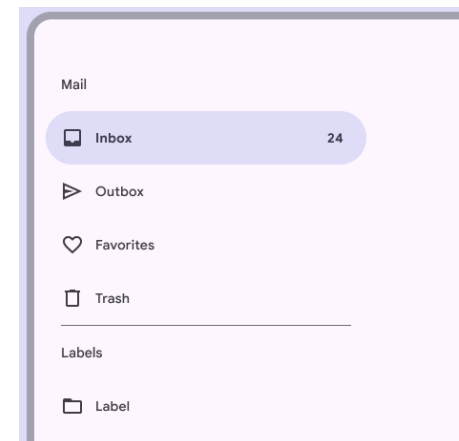
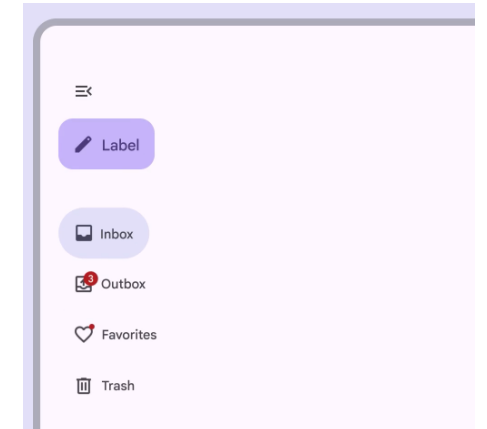
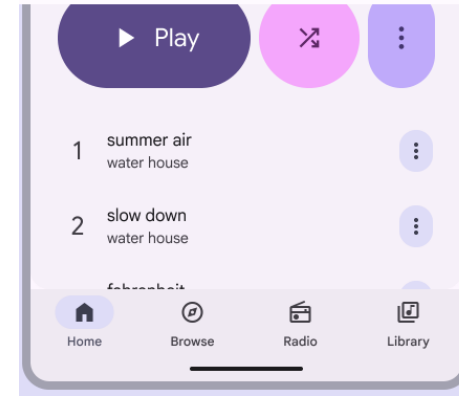


Material Components as UX Decision

➤ Navigation patterns:

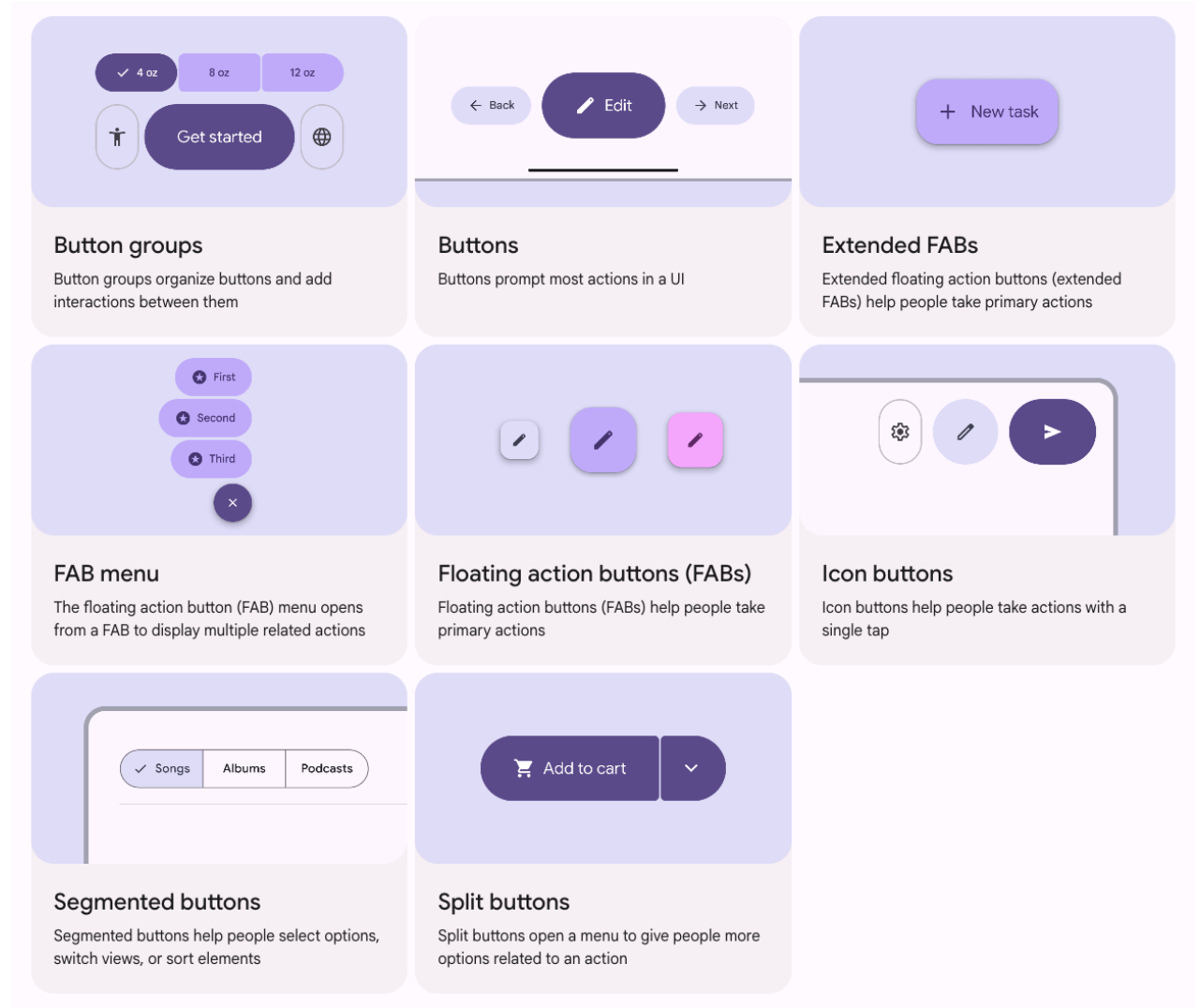
- Bottom navigation (few top sections)
- Navigation rail (larger screens)
- Drawer (complex systems)

➤ Material links navigation model to app size.



Material Components as UX Decision

- **Content organization:**
 - Cards for grouping
 - Lists for scanning
 - FAB for dominant actions
- Each has semantic meaning.



Activating Material in Flutter: Theming

- In Flutter, Material is activated providing a ThemeData to your MaterialApp through:

```
MaterialApp(  
  theme: ThemeData(  
    useMaterial3: true,  
    colorScheme: ColorScheme.fromSeed(seedColor: Colors.blue),  
  ),  
)
```

- Why this matters:
 - Color harmony is automatic
 - Components become consistent
 - Accessibility improves
 - You stop hardcoding colors

Note: if you do not provide a ThemeData, Material is activated by default but you won't have global control.

- **Advice:** Start global. Avoid styling per screen.

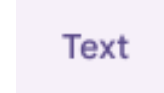
Buttons as UX Communication

➤ Different Buttons for different meanings

```
ElevatedButton(  
  onPressed: () {},  
  child: Text("Elevated"),  
)
```



```
TextButton(  
  onPressed: () {},  
  child: Text("Text"),  
)
```



➤ Button hierarchy allows to specify primary and secondary actions and put different emphasis.

➤ This allows also to express priorities.

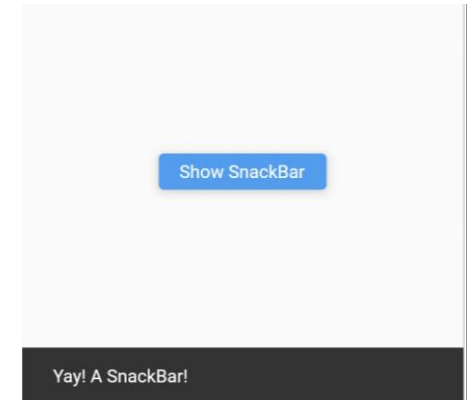
Feedback in Flutter

➤ Use the following to provide specific feedback.

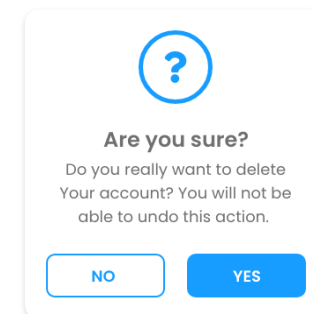
➤ Loading: `CircularProgressIndicator()`



➤ Confirmation: `ScaffoldMessenger.of(context).showSnackBar(SnackBar(content: Text("Yay! A SnackBar")),);`



➤ Critical action: `showDialog(...)`



Defining the Structure with Scaffold

➤ Material screens in Flutter use the following to define the basic structure of a screen:

- Scaffold
- AppBar
- NavigationBar
- FloatingActionButton

```
Scaffold(  
  appBar: AppBar(title: Text("Home")),  
  body: ListView(  
    padding: EdgeInsets.all(16),  
    children: [  
      Card(  
        child: ListTile(  
          title: Text("A text"),  
          subtitle: Text("A subtitle"),  
        ),  
      ],  
    ),  
  ),  
  floatingActionButton: FloatingActionButton(  
    onPressed: () {},  
    child: Icon(Icons.add),  
  ),  
);
```



Focus on the Scaffold

➤ Let's focus into the Scaffold Widget

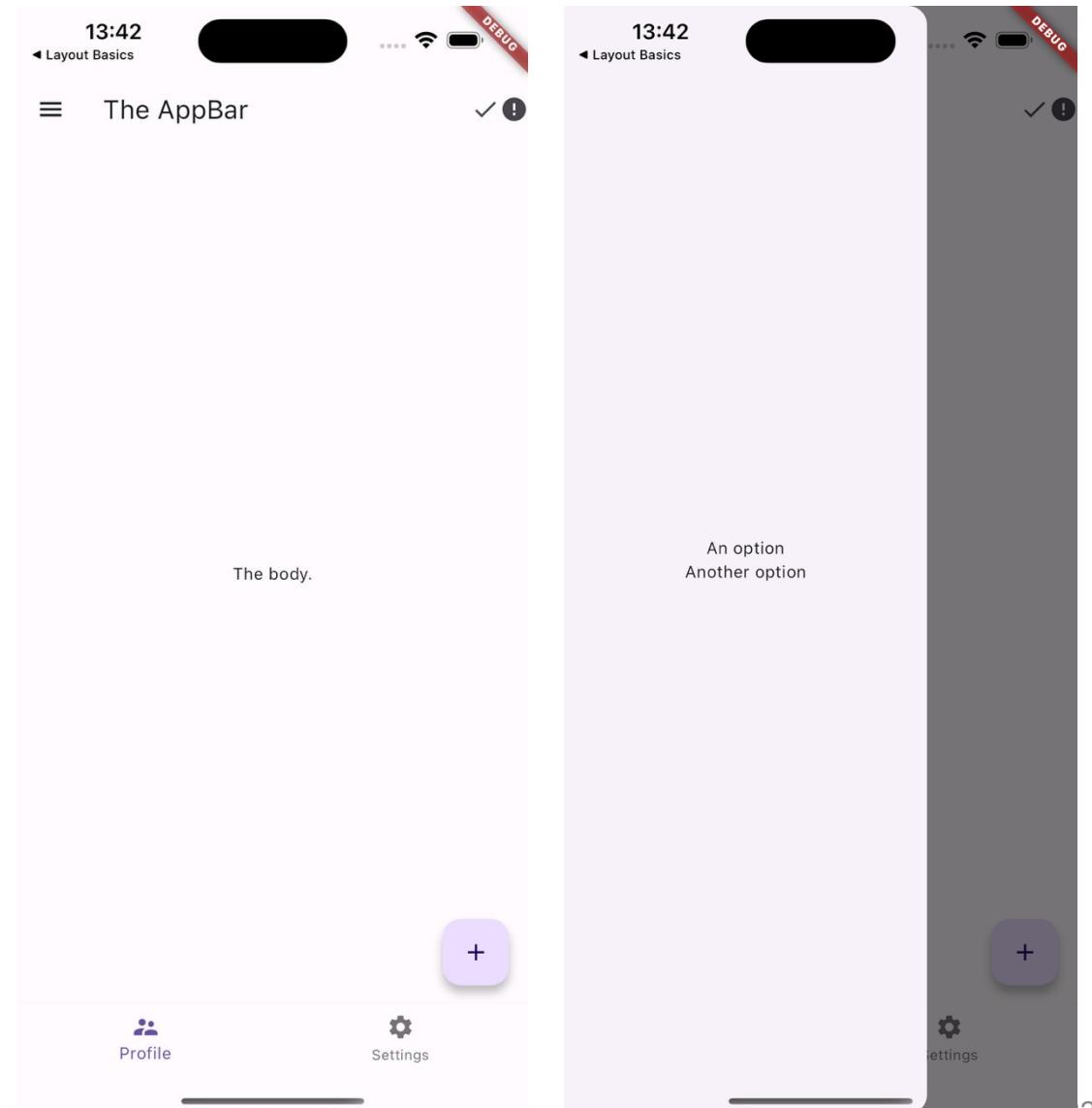
➤ Detailed info:
<https://api.flutter.dev/flutter/material/Scaffold-class.html>

```
//Constructor of Scaffold
const Scaffold({
  Key key,
  this.appBar,
  this.body,
  this.floatingActionButton,
  this.floatingActionButtonLocation,
  this.floatingActionButtonAnimator,
  this.persistentFooterButtons,
  this.drawer,
  this.endDrawer,
  this.bottomNavigationBar,
  this.bottomSheet,
  this.backgroundColor,
  this.resizeToAvoidBottomPadding = true,
  this.primary = true,
})
```

Focus on the Scaffold

//Constructor of Scaffold

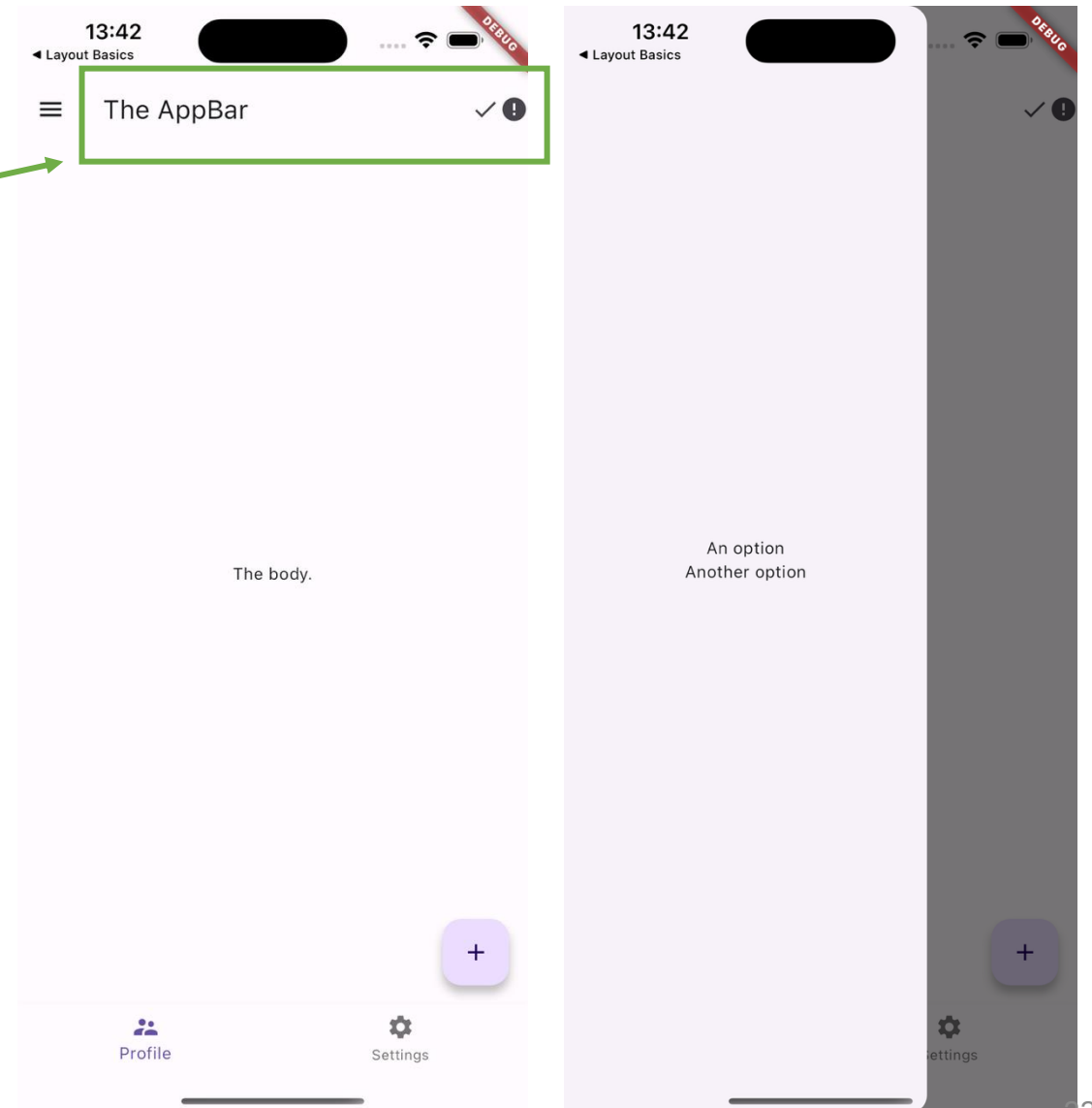
```
const Scaffold({  
  Key key,  
  this.appBar,  
  this.body,  
  this.floatingActionButton,  
  this.floatingActionButtonLocation,  
  this.floatingActionButtonAnimator,  
  this.persistentFooterButtons,  
  this.drawer,  
  this.endDrawer,  
  this.bottomNavigationBar,  
  this.bottomSheet,  
  this.backgroundColor,  
  this.resizeToAvoidBottomPadding = true,  
  this.primary = true,  
})
```



Focus on the Scaffold

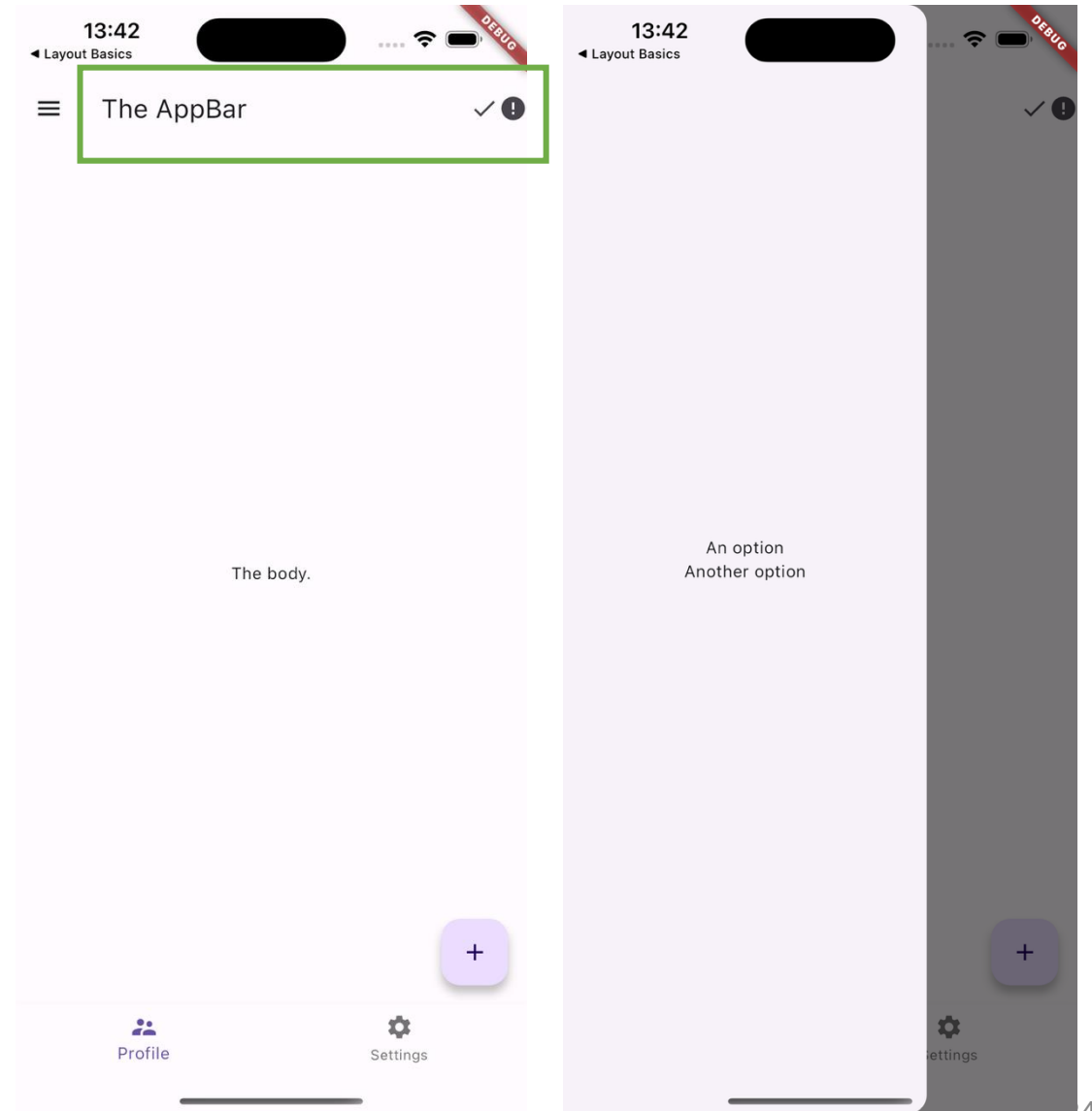
//Constructor of Scaffold

```
const Scaffold({  
  Key key,  
  this.appBar,  
  this.body,  
  this.floatingActionButton,  
  this.floatingActionButtonLocation,  
  this.floatingActionButtonAnimator,  
  this.persistentFooterButtons,  
  this.drawer,  
  this.endDrawer,  
  this.bottomNavigationBar,  
  this.bottomSheet,  
  this.backgroundColor,  
  this.resizeToAvoidBottomPadding = true,  
  this.primary = true,  
})
```



Focus on the Scaffold

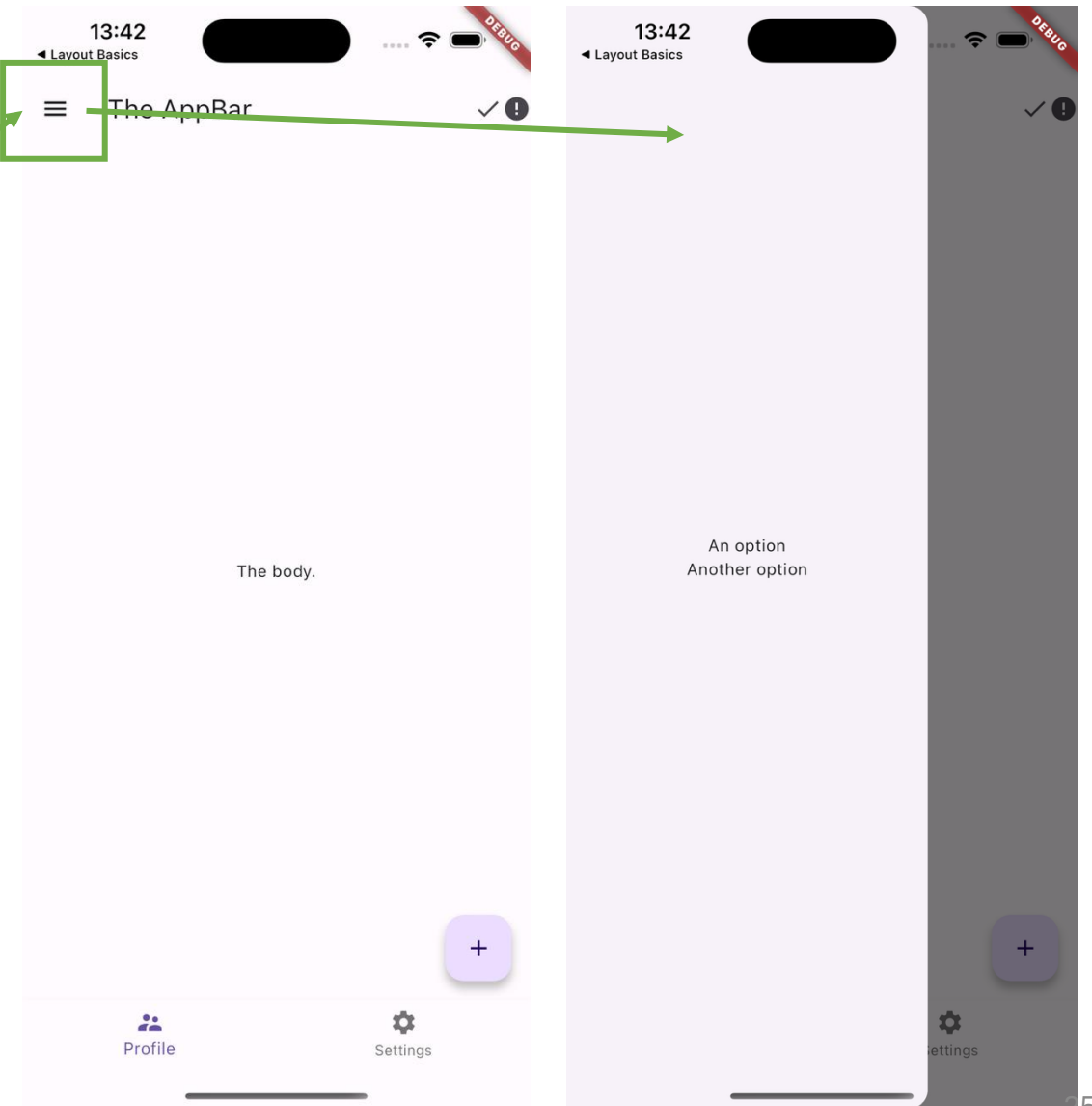
```
...  
home: Scaffold(  
  appBar: AppBar(  
    actions: [  
      Icon(Icons.done,),  
      Icon(Icons.error),  
    ],  
    title: Text('The AppBar'),  
  ),  
  ...  
),
```



Focus on the Scaffold

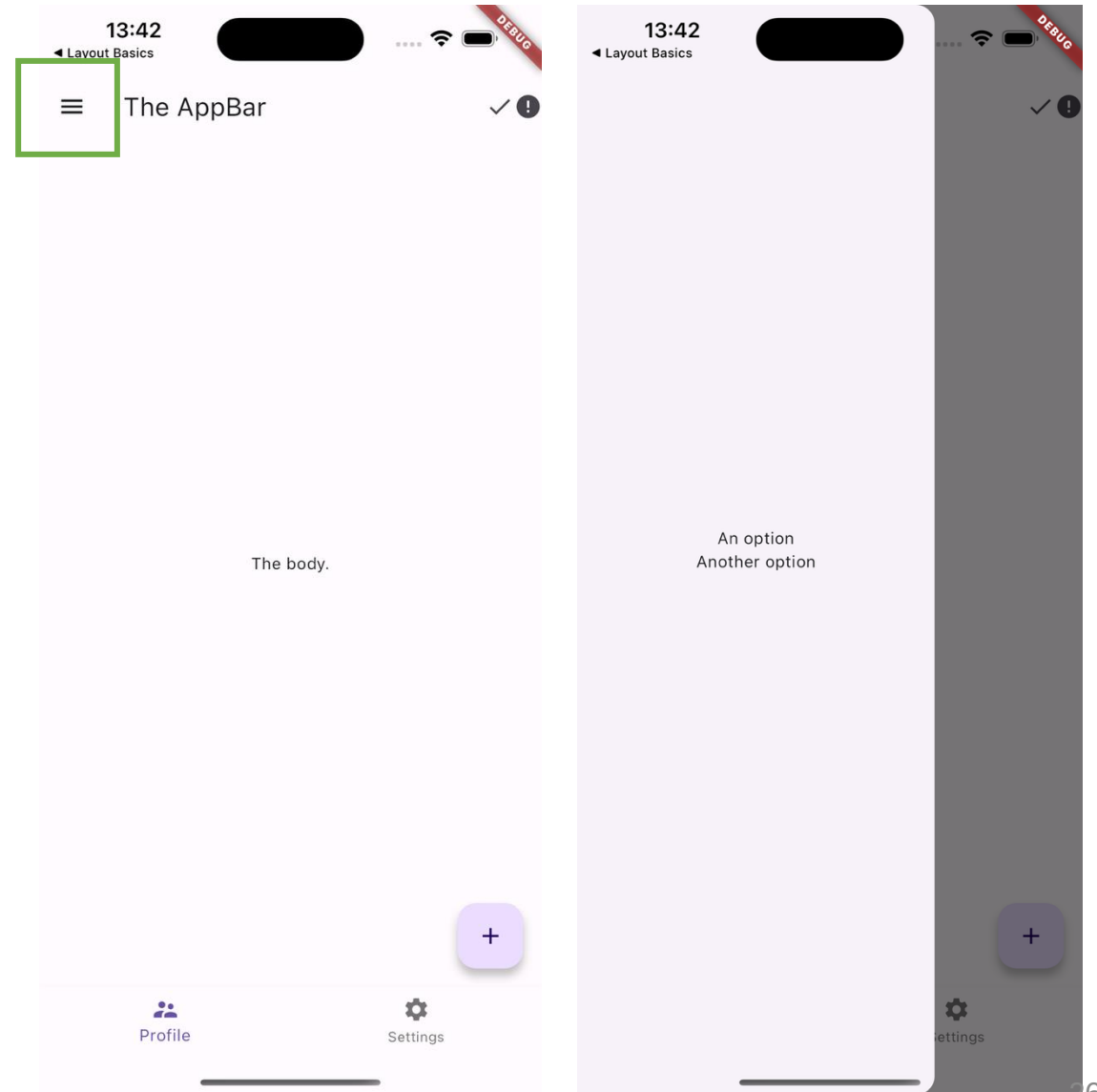
//Constructor of Scaffold

```
const Scaffold({  
  Key key,  
  this.appBar,  
  this.body,  
  this.floatingActionButton,  
  this.floatingActionButtonLocation,  
  this.floatingActionButtonAnimator,  
  this.persistentFooterButtons,  
  this.drawer,  
  this.endDrawer,  
  this.bottomNavigationBar,  
  this.bottomSheet,  
  this.backgroundColor,  
  this.resizeToAvoidBottomPadding = true,  
  this.primary = true,  
})
```



Focus on the Scaffold

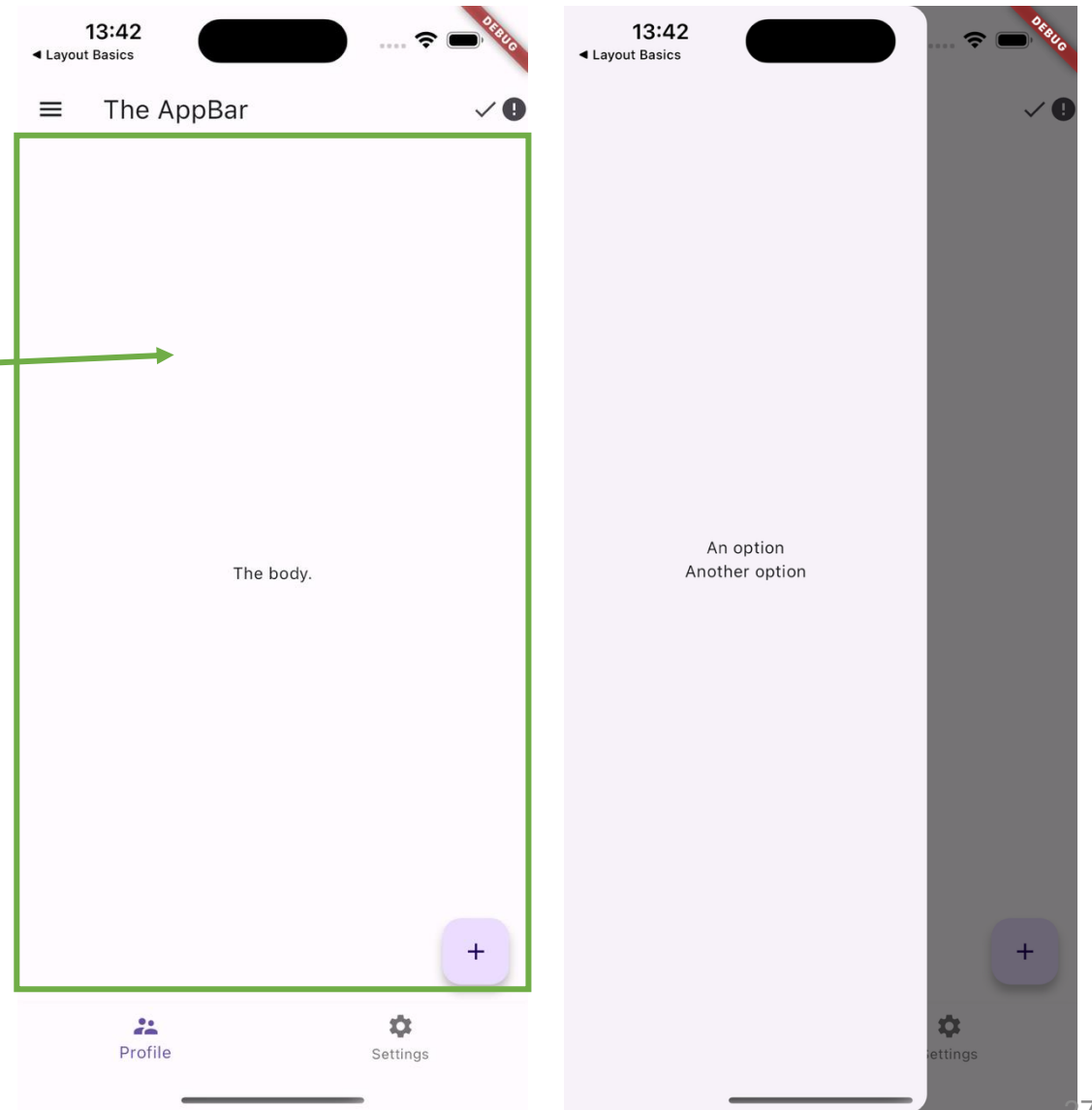
```
...
home: Scaffold(
  ...
  drawer: Drawer(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        Text('An option'),
        Text('Another option'),
      ],
    ),
  ),
  ...
```



Focus on the Scaffold

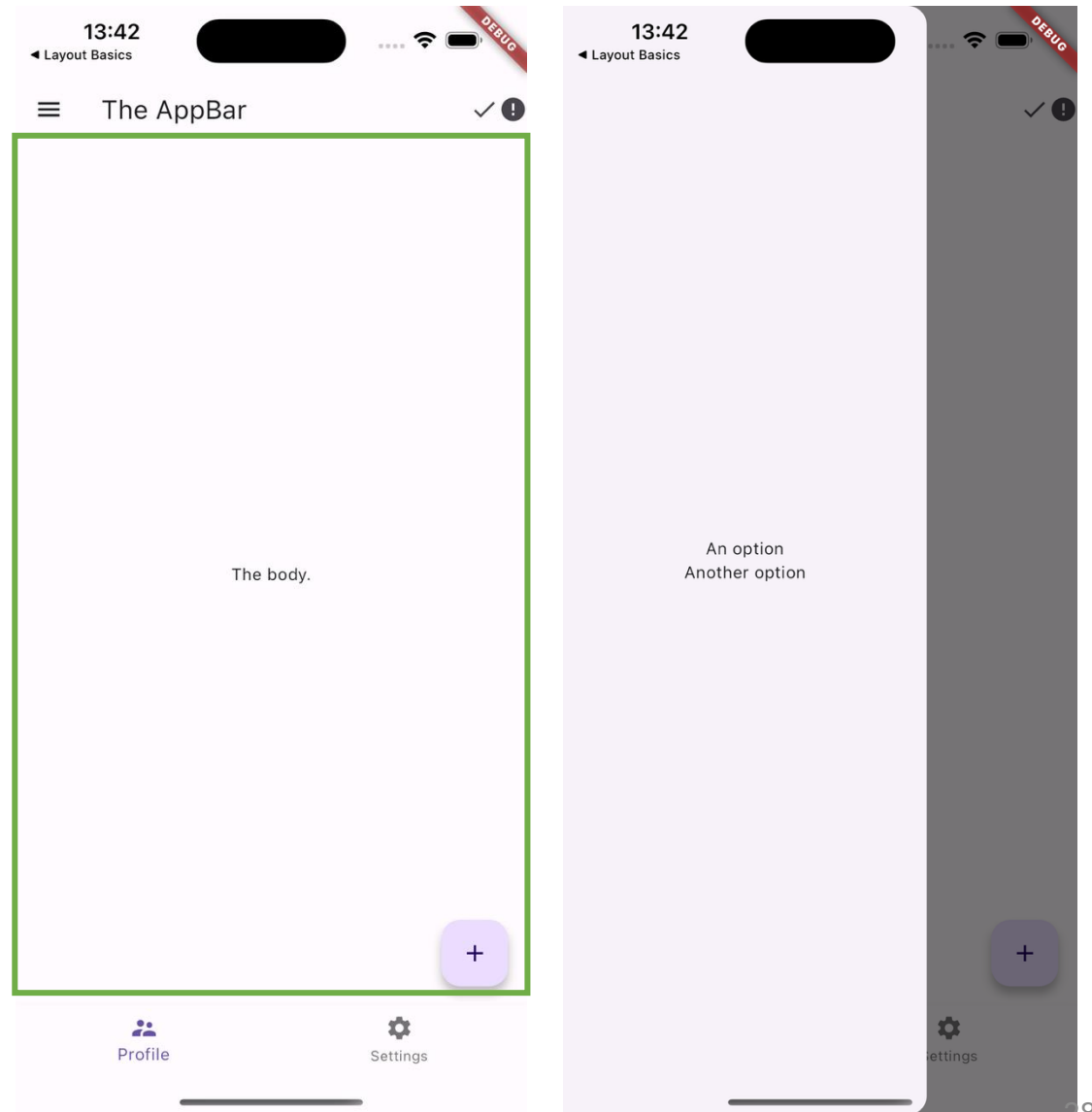
//Constructor of Scaffold

```
const Scaffold({  
  Key key,  
  this.appBar,  
  this.body,  
  this.floatingActionButton,  
  this.floatingActionButtonLocation,  
  this.floatingActionButtonAnimator,  
  this.persistentFooterButtons,  
  this.drawer,  
  this.endDrawer,  
  this.bottomNavigationBar,  
  this.bottomSheet,  
  this.backgroundColor,  
  this.resizeToAvoidBottomPadding = true,  
  this.primary = true,  
})
```



Focus on the Scaffold

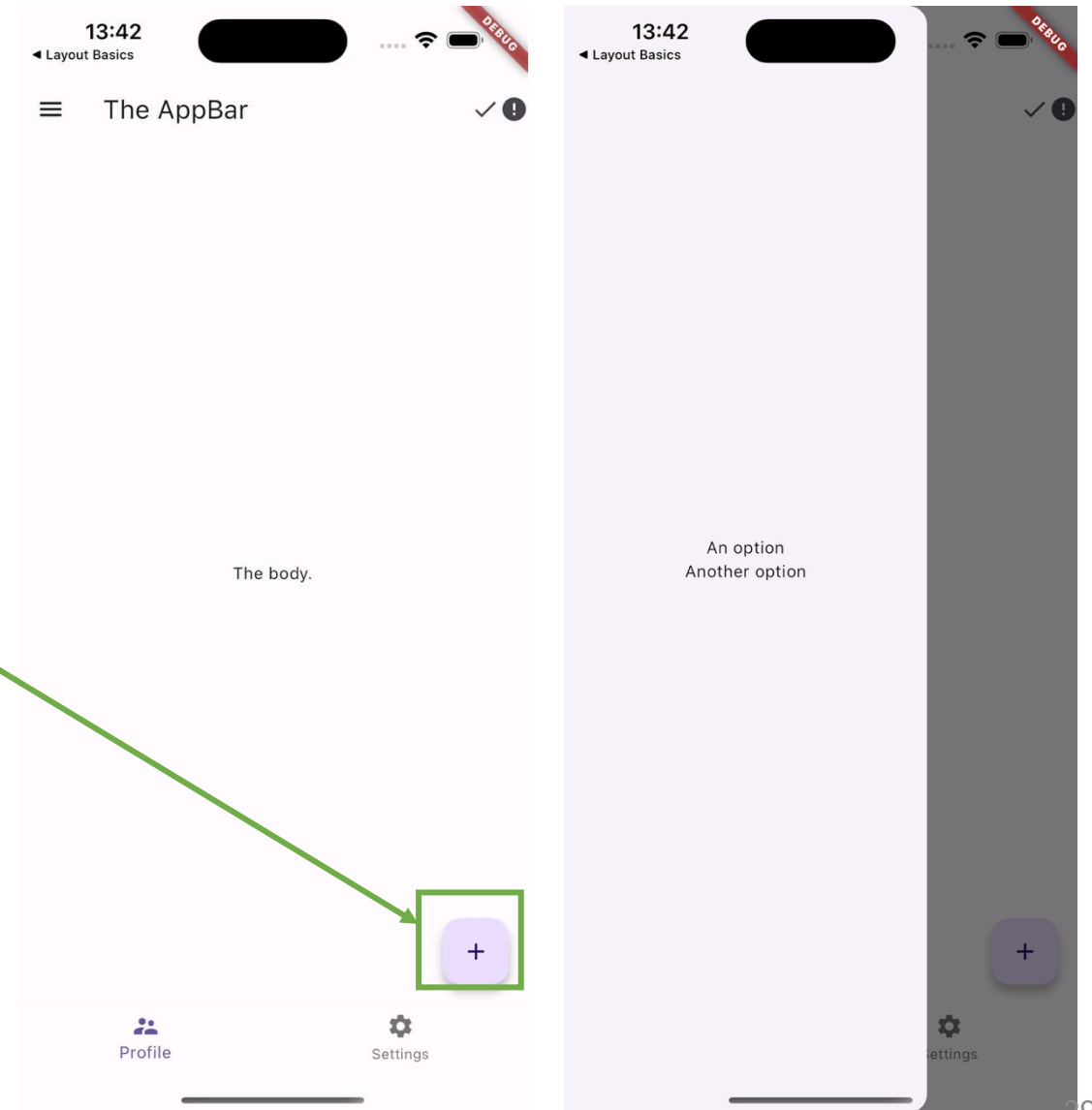
```
...  
home: Scaffold(  
  ...  
  body: Center(  
    child: Text('The body.'),  
  ),  
  ...  
)
```



Focus on the Scaffold

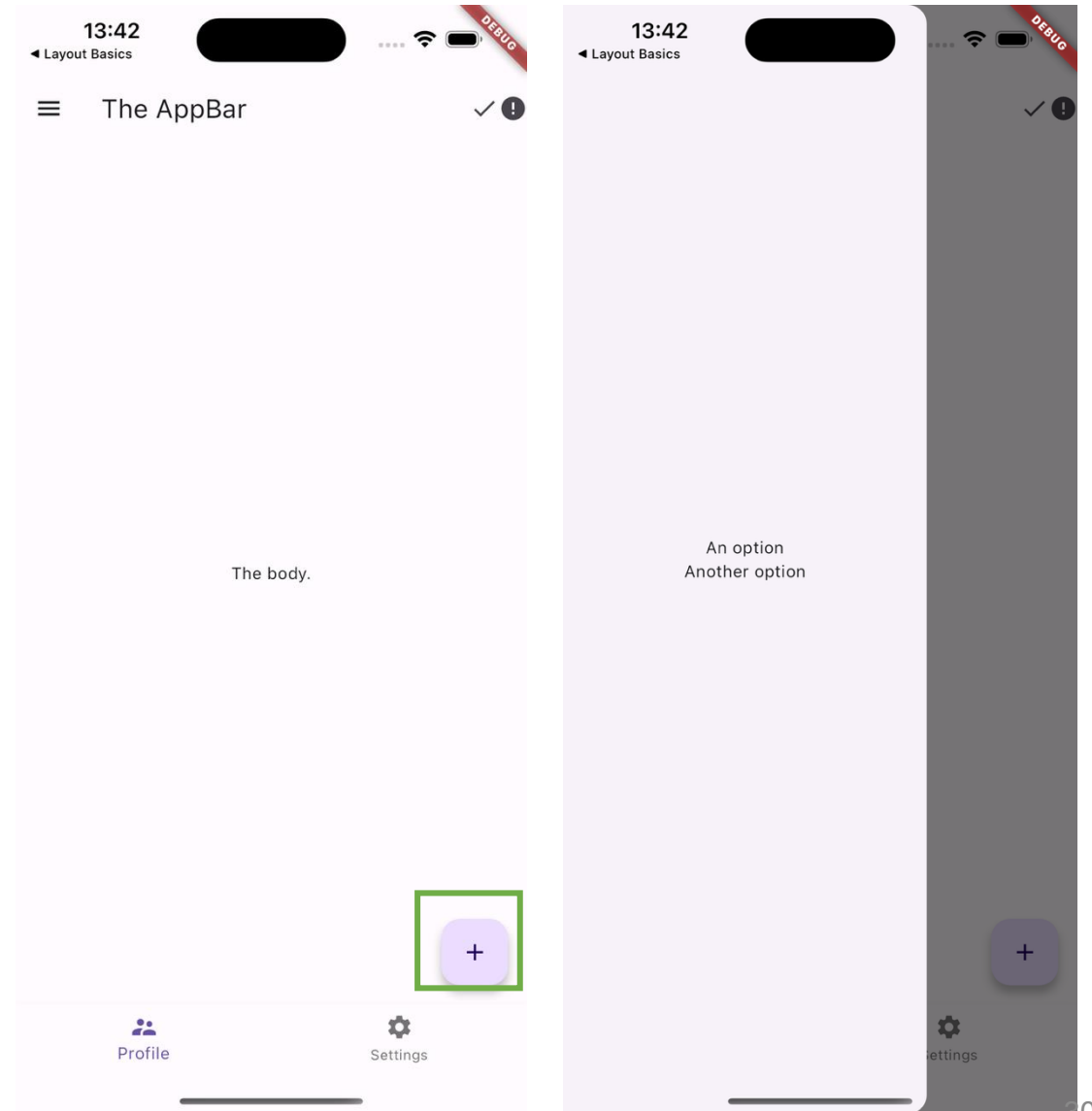
//Constructor of Scaffold

```
const Scaffold({  
  Key key,  
  this.appBar,  
  this.body,  
  this.floatingActionButton,  
  this.floatingActionButtonLocation,  
  this.floatingActionButtonAnimator,  
  this.persistentFooterButtons,  
  this.drawer,  
  this.endDrawer,  
  this.bottomNavigationBar,  
  this.bottomSheet,  
  this.backgroundColor,  
  this.resizeToAvoidBottomPadding = true,  
  this.primary = true,  
})
```



Focus on the Scaffold

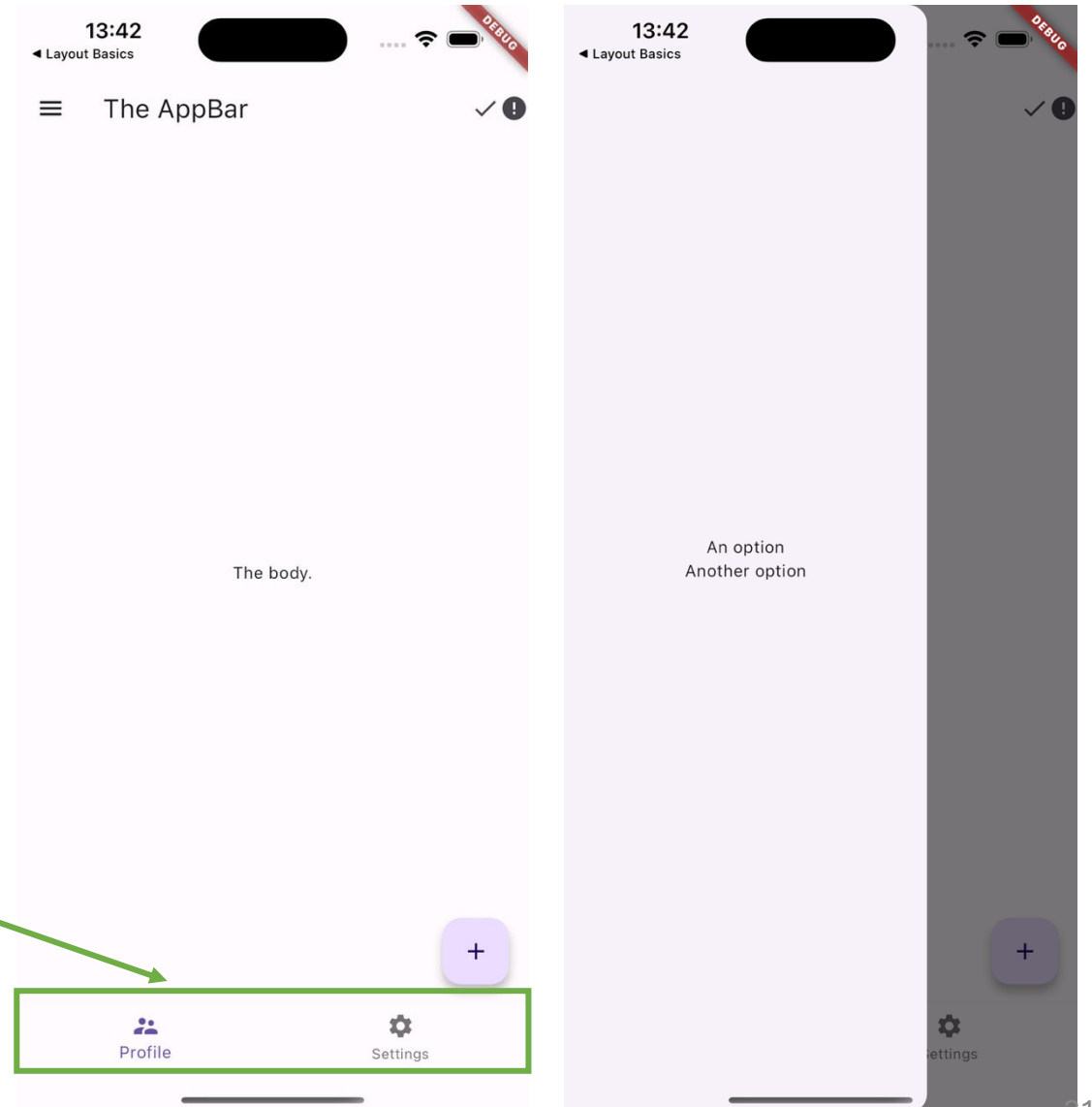
```
...  
home: Scaffold(  
  ...  
  floatingActionButton:  
    FloatingActionButton(  
      child: Icon(Icons.add),  
      onPressed: () {},  
    ),  
  ...  
)
```



Focus on the Scaffold

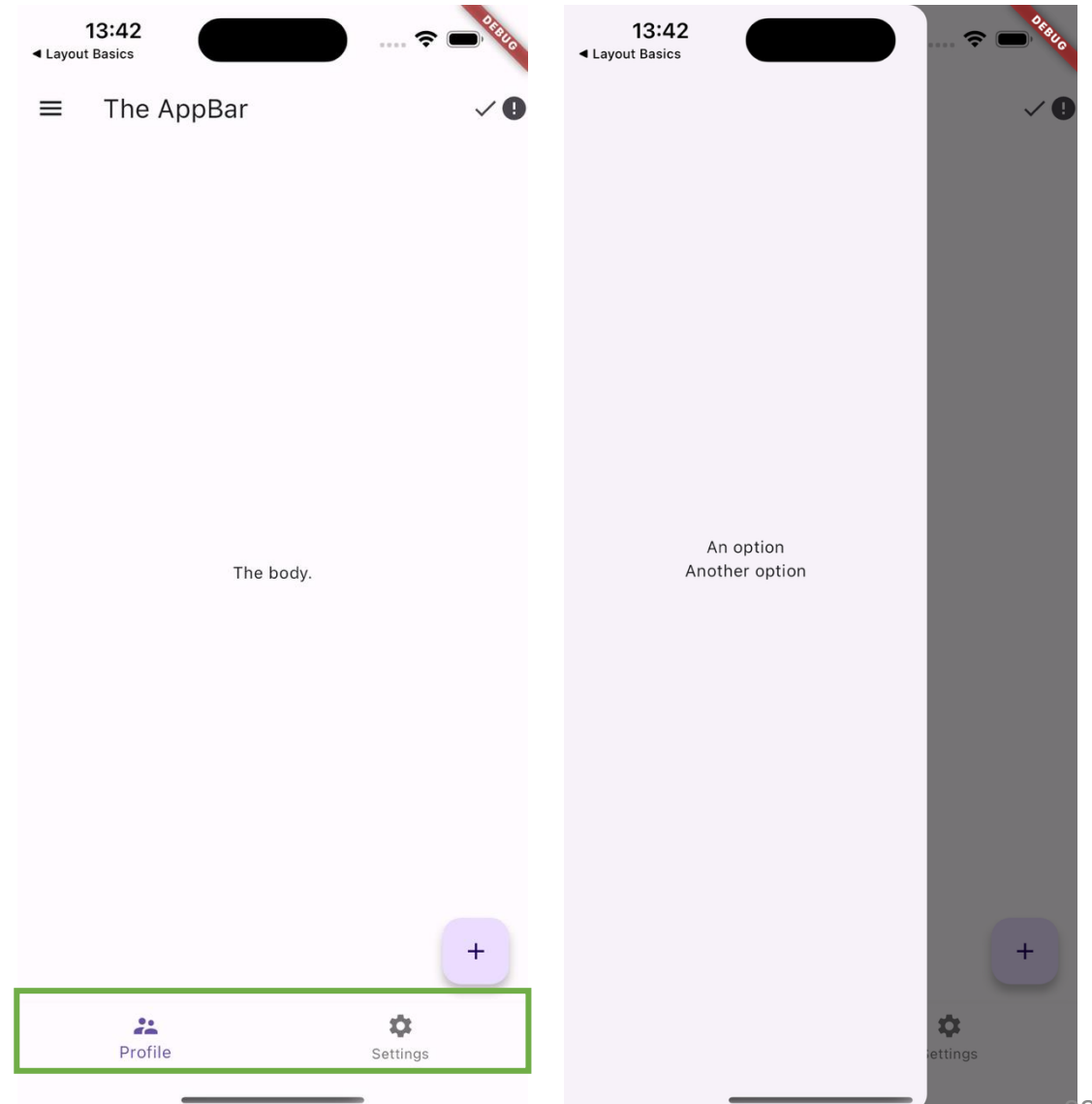
//Constructor of Scaffold

```
const Scaffold({  
  Key key,  
  this.appBar,  
  this.body,  
  this.floatingActionButton,  
  this.floatingActionButtonLocation,  
  this.floatingActionButtonAnimator,  
  this.persistentFooterButtons,  
  this.drawer,  
  this.endDrawer,  
  this.bottomNavigationBar,  
  this.bottomSheet,  
  this.backgroundColor,  
  this.resizeToAvoidBottomPadding = true,  
  this.primary = true,  
})
```



Focus on the Scaffold

```
...  
home: Scaffold(  
  ...  
  bottomNavigationBar:  
    BottomNavigationBar(  
      items: [  
        BottomNavigationBarItem(  
          icon: Icon(Icons.supervisor_account),  
          label: 'Profile',  
        ),  
        BottomNavigationBarItem(  
          icon: Icon(Icons.settings),  
          label: 'Settings',  
        )  
      ],  
    ),  
  ),  
),
```

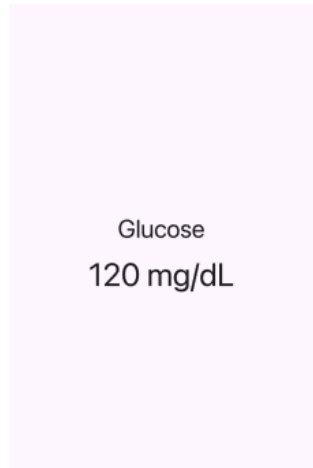


Grouping Widgets together

- Grouping is how we visually communicate relationships.
- Humans automatically assume:
 - Elements close together are related.
 - Elements inside the same container belong together.
 - Aligned elements form a logical group.
- If we don't group properly:
 - Screens feel chaotic
 - Users don't know what belongs together
 - Cognitive load increases
- Grouping reduces mental effort.

Column: The Vertical Grouping

- Use the Column Widget when:
- Elements belong together vertically
 - The user should read top-to-bottom
 - You are structuring a section



```
Column(  
  mainAxisAlignment: MainAxisAlignment.center,  
  children: [  
    Text(  
      "Glucose",  
      style: Theme.of(context).textTheme.titleLarge,  
    ),  
    SizedBox(height: 8),  
    Text(  
      "120 mg/dL",  
      style: Theme.of(context).textTheme.headlineMedium,  
    ),  
  ],  
)
```

Row: The Horizontal Grouping

➤ Use the Row Widget when:

- Items are at the same hierarchy level
- They represent related but parallel information
- You want side-by-side structure

Glucose

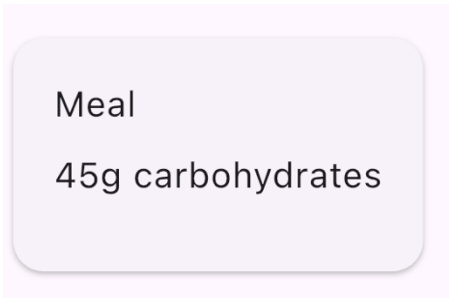
120 mg/dL

```
Row(  
  mainAxisAlignment: MainAxisAlignment.center,  
  children: [  
    Text(  
      "Glucose",  
      style: Theme.of(context).textTheme.titleLarge,  
    ),  
    SizedBox(height: 8),  
    Text(  
      "120 mg/dL",  
      style: Theme.of(context).textTheme.headlineMedium,  
    ),  
  ],  
)
```

Card: The Unit Grouping

➤ Use the Card Widget to communicate that:

- This content is a unit
- It belongs together
- It can be interacted with as one block



```
Card(  
  child: Padding(  
    padding: EdgeInsets.all(16),  
    child: Column(  
      crossAxisAlignment: CrossAxisAlignment.start,  
      children: [  
        Text("Meal"),  
        SizedBox(height: 8),  
        Text("45g carbohydrates"),  
      ],  
    ),  
  ),  
)
```

ListView

- Use ListView if you want communicate that:
 - These are similar items
 - They share the same structure
 - They can grow dynamically
- Conceptually they support scanning and pattern recognition in a UI.
- Detailed info:
<https://api.flutter.dev/flutter/widgets/ListView-class.html>



ListView

```
...
home: Scaffold(
  ...
  body: ListView(
    children: [
      ListTile(
        leading: Icon(Icons.settings),
        title: Text('ListTile 1 title'),
        subtitle: Text('ListTile 1 subtitle'),
        trailing: Icon(Icons.arrow_right),
      ),
      ListTile(
        leading: Icon(Icons.settings),
        title: Text('ListTile 2 title'),
        subtitle: Text('ListTile 2 subtitle'),
        trailing: Icon(Icons.arrow_right),
      ),
    ],
  ),
),
```

